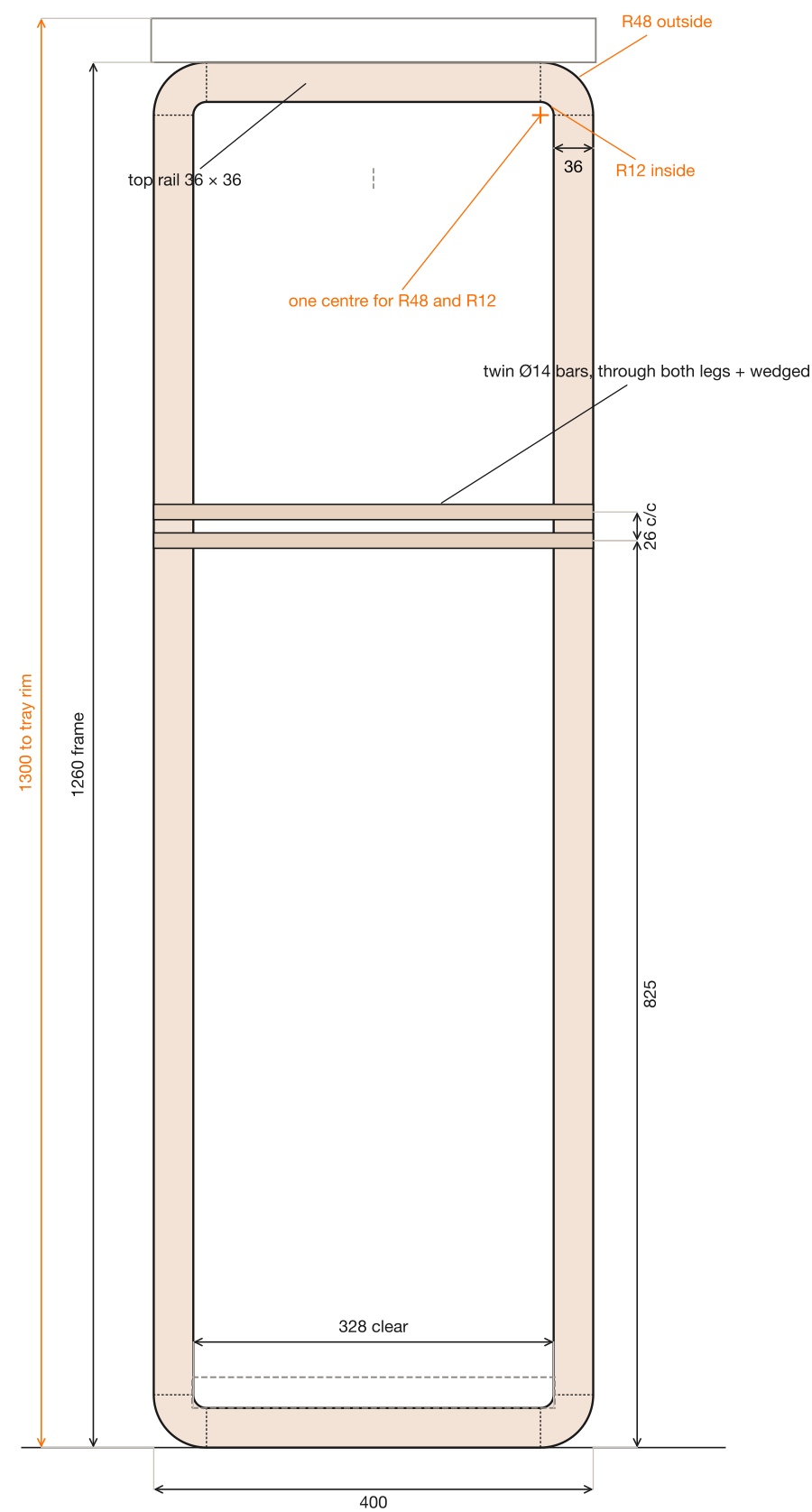
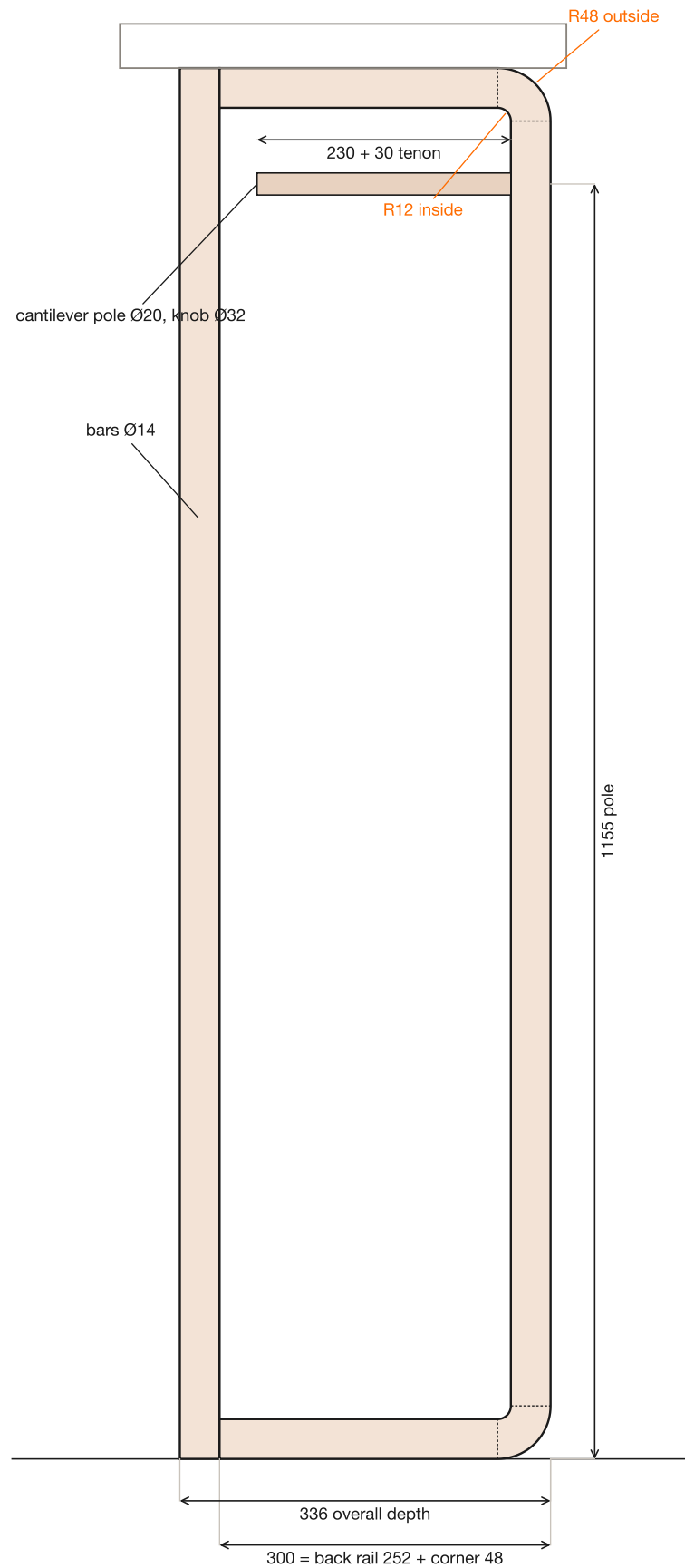


Valet stand — frame

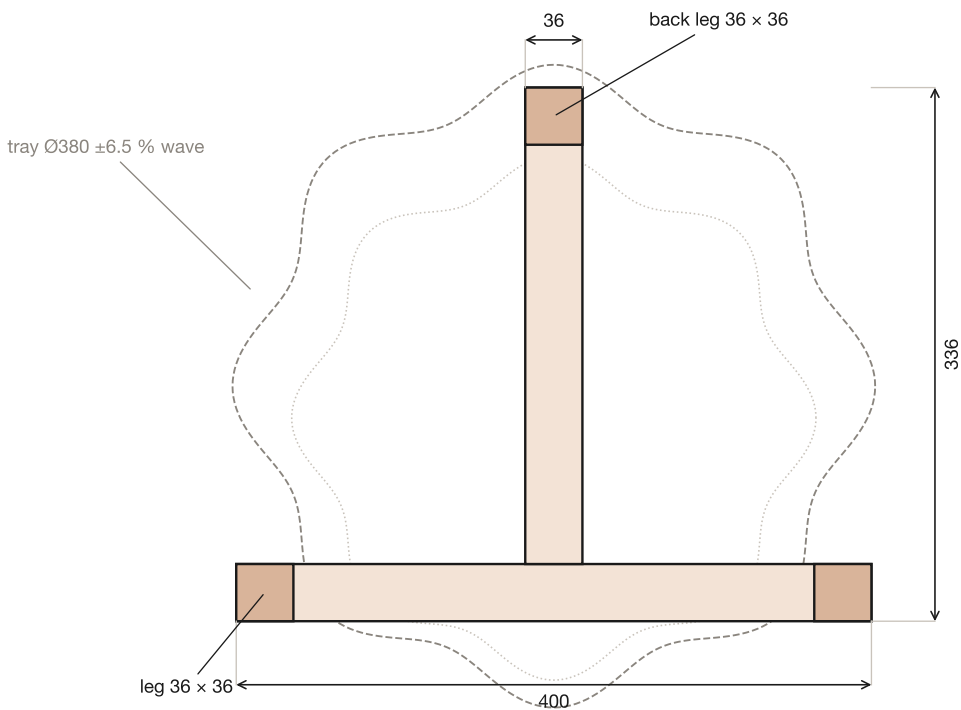
Front elevation · side elevation · plan · tray top 1300 mm · legs + rails 36 × 36 · corners R48 / R12, one centre · all dimensions in mm



FRONT ELEVATION



SIDE ELEVATION



PLAN (tray dashed)

- Frame stock: mahogany (sipo). Every member 36 × 36: legs, rails and corners.
- Corners are concentric (R48 / R12, one centre), so the band stays 36 wide through the bend.
 - Corners: mitred blocks glued on oversize, then routed to shape (sheet 3). Joints = dashed lines.
 - Every joint: 2× Ø8 × 60 dowels, 35 into the straight part, 25 into the corner.
 - Back rails into the back face of the front rails: 2× Ø8 × 60, 35 + 25.
 - All arrises R4 round-over, routed after glue-up so it runs unbroken over every joint.
 - Twin trouser bars stacked (as drawn): side by side would leave 4 mm of wall.

Cut list · order of work · tray top 1300 mm · legs + rails 36 × 36 · corners R48 / R12, one centre · all dimensions in mm

	PART	QTY	LENGTH	SECTION	WOOD	NOTES
A	Front leg	2	1164	36 × 36	Mahogany	straight, square ends; 2× Ø8 holes 36 deep in each end
D	Back leg	1	1164	36 × 36	Mahogany	as A
B	Front rails (top + bottom)	2	304	36 × 36	Mahogany	straight, square ends; 2× Ø8 holes 36 deep in each end
E	Back rails (top + bottom)	2	252	36 × 36	Mahogany	straight; front end into the back face of B, back end into a corner piece
K	Corner blocks	6	99.5 long edge	43 × 36	Mahogany	mitred 45° both ends like a picture frame (short edge 13.5); glued on oversize, routed to R48 / R12 after glue-up (sheet 3)
G	Trouser bars	2	400	Ø14	Mahogany	through both front legs at 825 / 851 from floor = 777 / 803 from leg bottom
H	Hanger pole	1	260	Ø20	Mahogany	30 mm into D at 1155 from floor = 1107 from leg bottom; knob Ø32 × 24
—	Joint dowels	28	60	Ø8	beech	14 joints × 2 (6 corners × 2 ends + 2 back rails); 35 + 25

Corner pieces: see sheet 3 (geometry, maths, blank layout, 1:1 template).

Order of work

1. Mill all mahogany to 36 mm; rip legs to 36 × 36 and cut them to 1164.
2. Cut the straight rails: 2× 304 (front) and 2× 252 (back), 36 × 36.
3. Rip one strip 43 × 36, ~450 long; mitre 6 blocks at 45°, long edge 99.5, flipping the strip between cuts.
4. Dowel holes Ø8: 36 deep in legs/rails, 26 deep in corners and the back face of B; bar holes through both front legs.
5. Glue the front ring (8 pieces + bars) and the back C (5 pieces); bandsaw corner waste, rout R48 / R12 with the sheet-3 template; join.
6. Drill the Ø20 pole hole into D at 1155 (1107 from its bottom), 30 deep; rout R4 on every arris; finish.

from a 43 x 36 strip, grain along the strip · orange dashed = finished shape after routing



Technical drawing of a square plate with two circular holes. The plate has a total width of 36 and a total height of 36. The top corners are rounded with a radius of R4. There are two circular holes, each with a diameter of 8 (Ø8), spaced 16 units apart vertically. The distance from the bottom edge to the center of the bottom hole is 10 units. The distance from the top edge to the center of the top hole is 16 units. The text "2x Ø8" is centered between the two holes.

Band $t = 36$ (legs = rails = corners)

Outside $R_o = 48$ · inside $R_i = R_o - t = 12$

Same centre C, so the band is 36 wide all round

$C = (152, 1212)$ from centreline / floor

Leg = $1260 - 2 \cdot 48 = 1164$

Front rail = $400 - 2 \cdot 48 = 304$

Back rail = $300 - 48 = 252$

Strip width = $48 - 12 \cdot \cos 45^\circ = 39.51 + 2 \cdot 1.75 = 43$

Block: long edge $2 \cdot 49.75 = 99.5$, short $2 \cdot 6.74 = 13.5$

Mitre face $43 / \cos 45^\circ = 60.8$; sticks out 22.4 past the leg

Arc lengths: top 75.4 · centre 47.1 · in 18.8

Grain at the ends crosses the band at 45° :

fibres $36 / \sin 45^\circ = 50.9$ long — no short grain

Dowel into corner: centre 30 from C, edge 34;

3 mm skin under R48: depth $\leq \sqrt{(45^2 - 34^2)} = 29.5$, use 25

Pole Ø20 at 1155: top 1165, leg-top holes

end at 1176: 11 mm clear

1. Rip one strip 43 x 36, ~450 long (same stock as the legs).
2. Mitre saw at 45°: 6 blocks, long edge 99.5. Flip the strip between cuts so the blocks nest (B).
3. Ø8 holes 26 deep in each mitre face, 20.5 from the short tip, 10 and 26 from the front. Legs/rails: dowel centre 18 in.
4. Glue the 8 pieces flat; block overhangs 22 outside, 2.5 inside.
5. Bandsaw the outside waste to ~2 mm off the line.
6. Rout with the template (A) + flush-trim bit, light passes, cut downhill with the 45° grain to avoid tear-out.
7. R4 round-over on all arrises, unbroken over the joints.

Inner R12 > bearing radius 6.35, so the bit follows it.

